

Vision and Need

Quantitative measures of movement can support discovery across disciplines, but the ability to collect such data remains constrained by costly systems and fragmented technical infrastructure. Commercial motion capture systems are priced for well-resourced institutions and as such, detailed movement measurement is confined to a relative few while clinics, studios and field sites that often observe movement have no practical methods to measure it. Advances in computer vision and the subsequent development of modern markerless motion capture systems have reduced this barrier, but these systems are often built for a particular domain or subfield. Groups working outside these fields (animal locomotion, atypical human movement and morphologies, cross-disciplinary work) must solve the same problem of implementing and integrating video synchronization, camera calibration, 2D pose estimation, 3D reconstruction, and visualization and analysis independently, resulting in repeated investment of time and public research funding in infrastructure that is difficult to reproduce, maintain, or transfer across domains.

There is therefore a need for shared infrastructure that is accessible to end users, adaptable to specialized research needs, and structured with components that communities can build on and contribute back to rather than repeatedly recreate. Infrastructure of this kind requires defined stewardship and governance as much as software.

The Free Motion Capture Project (FreeMoCap) is a free, open-source software maintained by The FreeMoCap Foundation that records and transforms synchronized videos from consumer grade webcams into research-grade 3D motion capture for decentralized scientific research, education, and training. This proposal requests Track 1 support to scope the organizational and infrastructure requirements needed to transition FreeMoCap from an open-source project maintained by a small core team into a secure and sustainable open-source ecosystem that is driven by the community. Ultimately, such infrastructure could reduce duplicated technical effort across research communities, broaden access to quantitative movement tools, and allow public investment in underlying methods and software to benefit a wider range of scientific, educational, and technical applications.

Current State of the Product and Community

FreeMoCap is currently distributed through GitHub and a standalone installer, with more than 10,000 GitHub stars, telemetry indicating over 15,000 users across 153 countries, and an active community forum with over 4,000 members. The software is organized as several interoperating repositories separated by responsibility: `skellycam` for USB-camera synchronization and recording, `skellytracker` as a modular interface for pose estimation, `skellyforge` to handle calibration, reconstruction, and post-processing of data, and `freemocap_blender_addon` to handle export to Blender.

The software has already been adopted across a range of research applications, including reaching and grasping biomechanics [cite], sagittal-plane gait mechanics under controlled postural adjustments [cite], multimodal corpus construction in computational linguistics [cite], and assessment of idle-animation believability [cite]. Independent validation studies have reported favorable agreement with marker-based motion capture for spatiotemporal gait parameters [cite], squat and countermovement-jump mechanics [cite], and functional gait assessments [cite]. Our own validation work has further characterized FreeMoCap for gait and balance assessment against marker-based motion capture and demonstrated that its modular architecture can incorporate multiple pose-

estimation backends. The same underlying workflow has also been extended beyond conventional human biomechanics, including motion capture of animal locomotion.

While user adoption continues to grow, it has outpaced the growth of the maintainer base. The majority of the architectural and operational knowledge needed to understand and modify the core pipeline lies within a small core development team, creating a sustainability bottleneck. The structures needed to onboard, support, and retain users and contributors, while distributing technical knowledge and responsibility beyond the core team, remain immature.

Track 1 will address this gap by: (1) identifying and characterizing FreeMoCap’s user and contributor communities and their needs; (2) reducing dependence on tacit core-team knowledge through documentation, educational resources, and clearer contribution pathways; and (3) developing and evaluating models for community-driven contribution, governance, maintenance, and technical stewardship.

Team Qualifications

The FreeMoCap team is well positioned to address this transition because it combines scientific and engineering expertise with more than five years of direct experience developing and maintaining the software, supporting its user community, and responding to the technical and organizational challenges that arise as the project grows. The FreeMoCap Foundation was incorporated as a nonprofit organization in XXXX to support long-term stewardship of the project, promoting its core values of advancing research, and promoting technological and scientific accessibility and education.

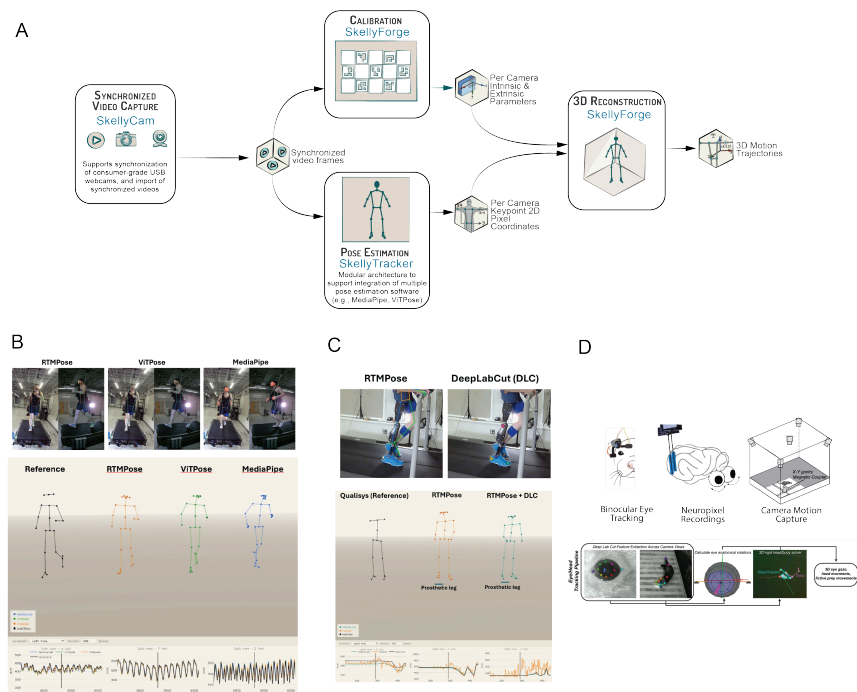


Figure 1: Placeholder. Rough sketch of what I’m thinking. A - graphic to do overview of subskellies and show how it works. B-D: examples of freemocap data, getting less and less ‘standard’ from left to right (i.e., human, then human with prosthetic, then non-human)

Ecosystem Discovery

Need for Innovation

Existing commercial (Theia3D, Move.ai, Rokoko, Meshcapade) and open-source (OpenCap, Pose2Sim, PosePipe) markerless motion capture tools demonstrate substantial demand, but differ in accessibility, licensing, extensibility and intended user communities. Domain-specific workflows continue to also be developed independently in research laboratories, suggesting that existing systems are still not sufficient to meet user needs. Track 1 scoping will look to test and understand the central FreeMoCap ecosystem strategy: a common, modular OSE can support both accessible general use and domain-specific extensions.

Necessities of an OSE approach

FreeMoCap's existing community provides early evidence that an OSE model can convert individual expertise into shared infrastructure benefiting the broader user community. Users have already become contributors in areas including animation, UI design, and pose estimation architecture, bringing domain expertise outside the core development team into the project. These transitions have occurred informally, however, without a consistent, defined pathway from user to contributor. Several contributors who have successfully entered the project have remained engaged over time, motivating Track 1 investigation of whether the bottleneck lies in entry rather than retention. Track 1 will develop and evaluate the organizational and technical structures needed to make participation more predictable, repeatable, and sustainable at scale.

Stakeholder discovery

Although telemetry points to over 15k users, our current understanding of the FreeMoCap community is biased towards users who voluntarily contact the core team or participate in community channels. Track 1 support will be used to combine analysis of the existing community, surveys, and the required I-Corps for PESOSE stakeholder-discovery process to systematically characterize current and potential users, contributors, and ecosystem partners. Interviews will span main stakeholder groups: researchers across application domains; educators and students using the software for instruction; creative and technical users in animation and related fields; current and prospective contributors; maintainers of existing and mature open-source projects.

Interviews with researchers across application domains will identify which components of the workflow require domain-specific adaption to research needs, producing a requirements map that distinguishes candidate core infrastructure from external extensions. Interviews with current and prospective contributors will identify where architectural knowledge, contribution processes, or unclear workflows prevent participation, informing a prioritized contributor-onboarding and documentation plan. Interviews with maintainers of mature open-source projects will examine community extension architecture and organization structures that can distribute development responsibility beyond the core team, producing a set of candidate models for evaluation in the Organization and Governance work.

Ecosystem Discovery will culminate in a segmented stakeholder and demand assessment, a prioritized map of user and contributor barriers, and a set of technical and organizational requirements that will guide the governance, security, community-building, and sustainability plans developed during Track 1.

Organization and Governance

The FreeMoCap Foundation was incorporated as a 501(c)(3) in XXXX and holds stewardship of the project's main repositories. Track 1 will be used to compare and evaluate models of how authority and responsibility should be distributed beyond the Foundation as the ecosystem grows.

Governance scoping

Track 1 will evaluate models used by comparable scientific and creative open-source projects, among them Python's PEP process, which separates a steering council holding final authority from a core team of trusted contributors and editors, against FreeMoCap's structure and scale. Interviews with maintainers of mature open-source projects during ecosystem discovery will be used to compare and evaluate different governance models.

[some other stuff here detailing more of the specifics of decision making/how authority is distributed, not sure what to put, will return to this]

Distributed infrastructure and community extensions

FreeMocap is currently organized as a series of repositories separated by responsibility, and this modular architecture provides an opportunity of separate shared infrastructure of motion capture from domain-specific elements required by particular users. For example, the `skellytracker` interface defines a pathway to implement alternative pose-estimation models into the broader motion capture workflow. This is particularly important as the models commonly used in markerless motion capture software are trained on standard human datasets and do not generalize to atypical morphologies, nonhuman subjects, or other specialized use cases, leaving each group to recreate a motion capture workflow for their use independently.

A community extension system would allow that integration work to be completed once and then reused. A researcher who has integrated a model best suited to a particular population could make it available to others facing the same need, while a computer vision researcher publishing a new method could translate their research to a much broader audience who might otherwise lack the technical knowledge to implement it. In this model, FreeMoCap would maintain the shared workflow, while domain experts could contribute specialized capabilities through defined interfaces.

Support for this model requires clear organization, governance and security practices. Track 1 will examine what review and quality standards extensions should meet, how approved extensions should be packaged and distributed, and how maintenance responsibilities should be divided between extension maintainers and core developers. These decisions will be informed by interviews conducted during ecosystem discovery with maintainers of mature open-source projects that already support community extension systems, as well as FreeMoCap users and researchers that can help define the documentation, interfaces, and technical knowledge required to develop and use such extensions. This work will produce a set of contribution, maintenance and review patterns that can guide not only pose-estimation extensions, but other community developed capabilities as well, such as biomechanical analyses.

Licensing

The FreeMoCap software and its component repositories are licensed under the AGPL. This choice reflects the central premise of the proposed ecosystem: improvements to publicly funded infrastructure are most valuable when they return to the broader community. The FreeMoCap Foundation also retains the ability to offer alternative licensing terms for organization whose intended use is incompatible with the AGPL, providing a potential sustainability mechanism -

keeping the OSE freely available while commercial licensing can generate resources for maintenance and stewardship.

Long-term Sustainability

Discussions with contributors and experienced open-source maintainers during Ecosystem Discovery will shape governance structures that reduce dependence on a small core team, allowing the ecosystem to be community-driven while remaining stewarded by the FreeMoCap Foundation. In parallel, the Foundation will evaluate a diversified model for financial sustainability, including alternative commercial licensing, standardized hardware kits, training and support services, institutional partnerships, and continued grant funding. Ultimately, we aim for sustainability to come from diversification, with an ecosystem that does not depend on a single maintainer, and an organization that does not depend on a single source of support.

Risk Analysis and Security Plan

The FreeMoCap workflow is entirely run on a user's own computer. Therefore, all generated data belongs solely to that user and is never passed through any cloud-based or other remote software, which is a substantial advantage for user privacy, researcher privacy with human subjects, and any clinical work.

FreeMoCap depends on a stack of computer vision and machine learning libraries. The community extension system proposed above would also introduce additional third-party code. Track 1 will assess the software against the best practices provided by the Open Source Security Foundation and will scope the review and approval controls needed to guard against malicious or inadvertently unsafe contributions for an extension system.

Data Quality

Scientific investment and trust must have the basis of scientific data quality. Changes to pose-estimation models, calibration, reconstruction, filtering, or other pipeline components can alter scientific outputs even when the software continues to function correctly. Track 1 will scope diagnostic, testing, and benchmarking infrastructure for detecting such changes, including comparison against reference datasets, documentation of expected performance, and regression testing across software versions and supported configurations. We will also define the metadata needed to trace an output dataset to the software versions, models, processing parameters, and other relevant configuration choices used to produce it.

[input ISO/Imatest collaboration here potentially]

Community Building

The FreeMoCap Foundation maintains an active Discord community of over 3,000 members, structured to allow users to ask and answer questions without relying on the core development team. The core team also hosts a weekly community call that provides a regular venue for project updates, user feedback, technical discussion, and interaction with potential contributors. These activities provide an established foundation for community engagement. However, rapid FreeMoCap user growth in 2026 (Figure 2) increased the need for educational and engagement models needed to support a substantially larger community.

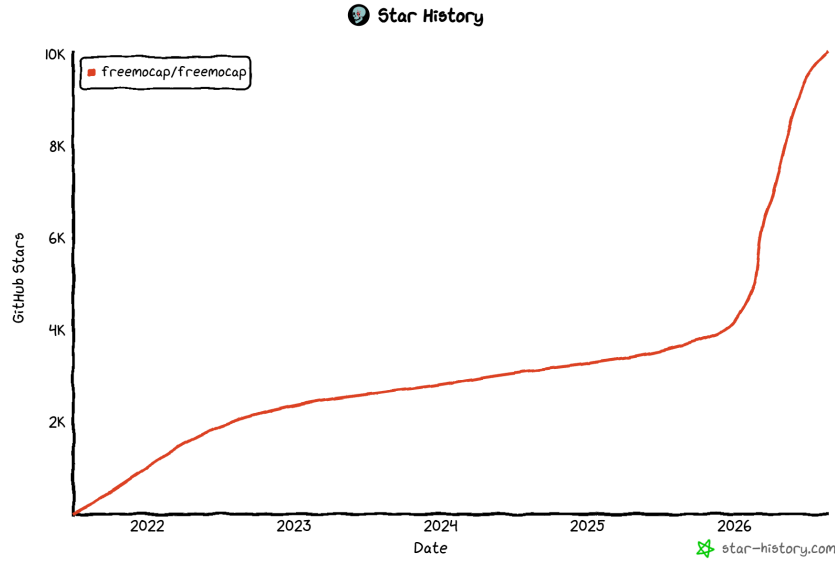


Figure 2: Github star chart

This work is also continuous with the contributor problem examined in Ecosystem Discovery. Educational materials are not only user support, they are also mechanisms by which to decentralize architectural knowledge of the FreeMoCap software and support contributor entry into the pipeline.

Track 1 will pilot scalable approaches to user education and community participation. These include hands-on classes and workshops developed with partner organizations, including Artisan’s Asylum and The Possible Zone, as well as the creation of tutorials, guided projects, and other educational resources that can be reused by the broader community. These pilots will identify which formats are most effective for different audiences, where learners encounter technical and knowledge barriers, and what materials can be made available at scale for for instruction.

Track 1 will also scope mechanisms for community contribution and participation that do not require coding. These may include calls for community-created artwork and visualizations, challenges, shared examples, and other participatory activities that allow users to contribute domain expertise and creative work to the broader ecosystem. These activities will test whether visible, low-barrier forms of participation can deepen engagement and create additional pathways toward sustained contribution.

The resulting materials will inform a scalable community-education model of documentation, curricula, tutorials and contribution pathways that can support a growing user base. Longer-term, these pilots will move towards establishing and hosting a larger community meeting or conference consisting of users, developers, educators, and researchers making use of the software, with potential support from a subsequent Track 2 application.

Milestones and Evaluation Plan

Qtr	Milestone / Deliverable	Evaluation
Q1	Complete I-Corps for PESOSE; establish baseline ecosystem metrics	Cohort completion confirmed by I-Corps instructors

Qtr	Milestone / Deliverable	Evaluation
Q1–Q2	Stakeholder discovery; prioritized user and contributor barrier map	Coverage of target stakeholder groups and recurring barriers confirmed by interviewed participants
Q2	Develop candidate governance, contributor, and stewardship models	Compared against discovery findings and reviewed by experienced OSE maintainers
Q2–Q3	Define core-versus-extension model and contribution, maintenance, licensing, and review requirements	Validated against the extension types identified in discovery; reviewed by maintainers of projects operating extension systems
Q3	Security and data-quality scoping; prototype automated benchmarking and regression suite	Suite detects known differences in output across trackers and versions; risks mapped to controls and reviewed externally
Q3	Pilot education and onboarding approaches with partner organizations	Formats assessed through participant feedback and observed barriers
Q4	Integrate findings into final governance, sustainability, community, and implementation roadmap	Governance model and contribution process documented at a level of detail sufficient for someone outside the core team to follow without consultation

Evaluation

Track 1 success will be evaluated by: 1. Documentation of governance roles, contributor pathways, and software architecture, assessed by whether contributors outside the core team can act on them without consultation; 2. Educational and onboarding materials, assessed through participant outcomes in pilot sessions; 3. A framework for developing, maintaining and reviewing community extensions, assessed through limited pilot implementations of the proposed extension process. Each will be revised in response to stakeholder interviews before incorporation into the final ecosystem plan.

Broader Impacts

Quantitative movement analysis is currently organized such that a researcher who needs to track something outside the scope of existing tools must build their own pipeline. These pipelines rarely translate beyond the environment they were created in. With the establishment of a sustainable, secure ecosystem, an adaptation built once can become available to everyone, increasing the return on public investment in scientific software. FreeMoCap reaches a multitude of communities outside of just biomechanics. By connecting this communities within a shared framework, we can create pathways for the translation of expertise. A contribution made by an animator can improve software for a clinician; a method published by a computer vision researcher can reach users who may have never implemented it themselves.

Realizing this process requires more than openly available software. A community-driven ecosystem depends on the distribution of knowledge. The educational activities piloted under this project will help develop and test the materials needed to help users with different technical backgrounds enter the ecosystem and progress toward independent use, and if desired, contribution. By

pairing extensible technical infrastructure with accessible pathways for learning and participation, FreeMoCap can broaden not only who is able to use quantitative movement tools, but also who is able to help shape and extend them

Results from Prior NSF Support

Award: TODO. **Amount:** TODO. **Period:** TODO.

Title: TODO

Intellectual Merit

TODO

Broader Impacts

TODO

Publications

No publications were produced under this award.

Research Products and Availability

TODO — software counts, and PAPPG names it explicitly